

```
In [2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
from sqlalchemy import create_engine
from scipy.stats import ttest_ind
import scipy.stats as stats

warnings.filterwarnings('ignore')
```

```
In [3]: # Creating MySQL database connection
engine = create_engine(
    "mysql+pymysql://root:Tvs%405755@localhost:3306/inventory"
)
# Fetching vendor summary data
df = pd.read_sql_query(
    "SELECT * FROM vendor_sales_summary",
    engine
)
df.head()
```

Out[3]:

	VendorNumber	VendorName	Brand	Description	PurchasePrice	ActualPrice	Volume
0	1128	BROWN-FORMAN CORP	1233	Jack Daniels No 7 Black	26.27	36.99	1750.0
1	4425	MARTIGNETTI COMPANIES	3405	Tito's Handmade Vodka	23.19	28.99	1750.0
2	17035	PERNOD RICARD USA	8068	Absolut 80 Proof	18.24	24.99	1750.0
3	3960	DIAGEO NORTH AMERICA INC	4261	Capt Morgan Spiced Rum	16.17	22.99	1750.0
4	3960	DIAGEO NORTH AMERICA INC	3545	Ketel One Vodka	21.89	29.99	1750.0

```
In [5]: #summary statistics
df.describe().T
```

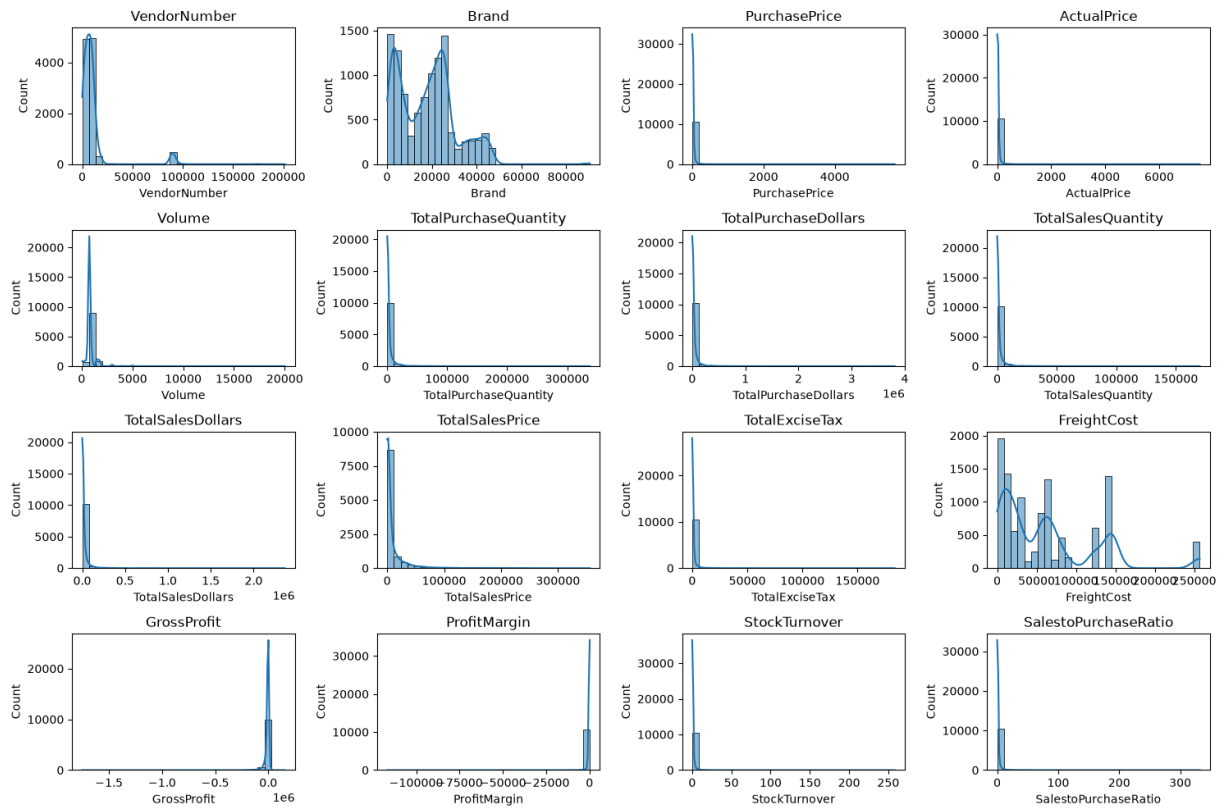
Out[5]:

	count	mean	std	min	25%
VendorNumber	10693.0	10649.892921	18752.805318	2.000000e+00	3951.000000
Brand	10693.0	18037.744319	12662.525384	5.800000e+01	5789.000000
PurchasePrice	10693.0	24.383023	109.264519	0.000000e+00	6.840000
ActualPrice	10693.0	35.640338	148.239484	0.000000e+00	10.990000
Volume	10693.0	847.351445	664.278813	5.000000e+01	750.000000
TotalPurchaseQuantity	10693.0	3140.781539	11094.573249	1.000000e+00	36.000000
TotalPurchaseDollars	10693.0	30103.877820	123062.388765	0.000000e+00	453.180000
TotalSalesQuantity	10693.0	1445.156457	5318.150983	0.000000e+00	9.000000
TotalSalesDollars	10693.0	19607.607876	77820.999310	0.000000e+00	174.500000
TotalSalesPrice	10693.0	9068.547549	22752.516039	0.000000e+00	47.760000
TotalExciseTax	10693.0	841.369223	5224.741572	0.000000e+00	1.350000
FreightCost	10693.0	61428.920862	60937.665572	9.000000e-02	14069.870000
GrossProfit	10693.0	-10496.269944	47726.667174	-1.751844e+06	-7368.840000
ProfitMargin	10692.0	-254.821488	2790.453158	-1.167031e+05	-65.337876
StockTurnover	10693.0	1.048483	4.789244	0.000000e+00	0.191489
SalestoPurchaseRatio	10692.0	1.533478	6.646334	0.000000e+00	0.278700

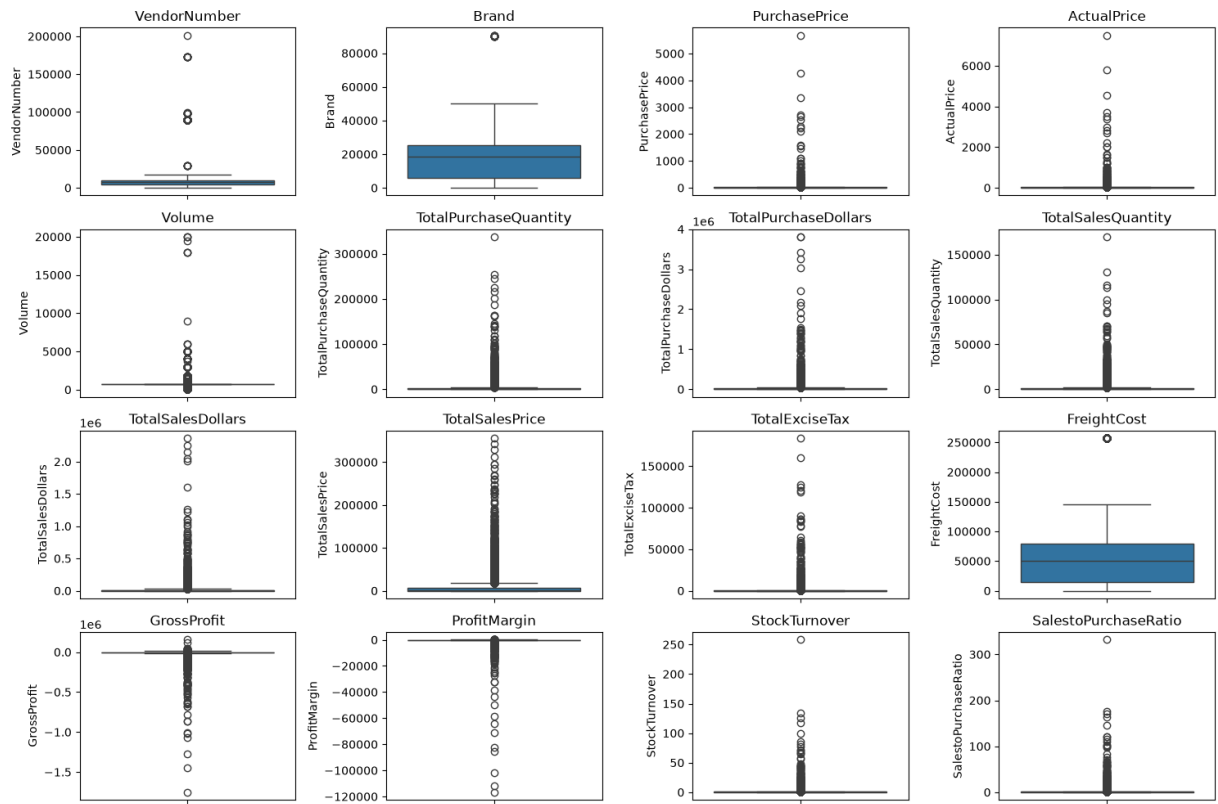


In [6]:

```
# Distribution Plots for Numerical Columns
numerical_cols = df.select_dtypes(include=np.number).columns
plt.figure(figsize=(15, 10))
for i, col in enumerate(numerical_cols):
    plt.subplot(4, 4, i + 1)      # Creates a 4 x 4 grid of subplots
    sns.histplot(df[col], kde=True, bins=30)
    plt.title(col)
plt.tight_layout()
plt.show()
```



```
In [7]: # Box Plots for Numerical Columns
numerical_cols = df.select_dtypes(include=np.number).columns
plt.figure(figsize=(15, 10))
for i, col in enumerate(numerical_cols):
    plt.subplot(4, 4, i + 1)      # Creates a 4 x 4 grid of subplots
    sns.boxplot(y=df[col])
    plt.title(col)
plt.tight_layout()
plt.show()
```



```
In [11]: # filter the data by removing inconsistencies
df = pd.read_sql_query("""
SELECT *
FROM vendor_sales_summary
WHERE GrossProfit > 0
AND ProfitMargin > 0
AND TotalSalesQuantity > 0
""", engine)
df
```

Out[11]:

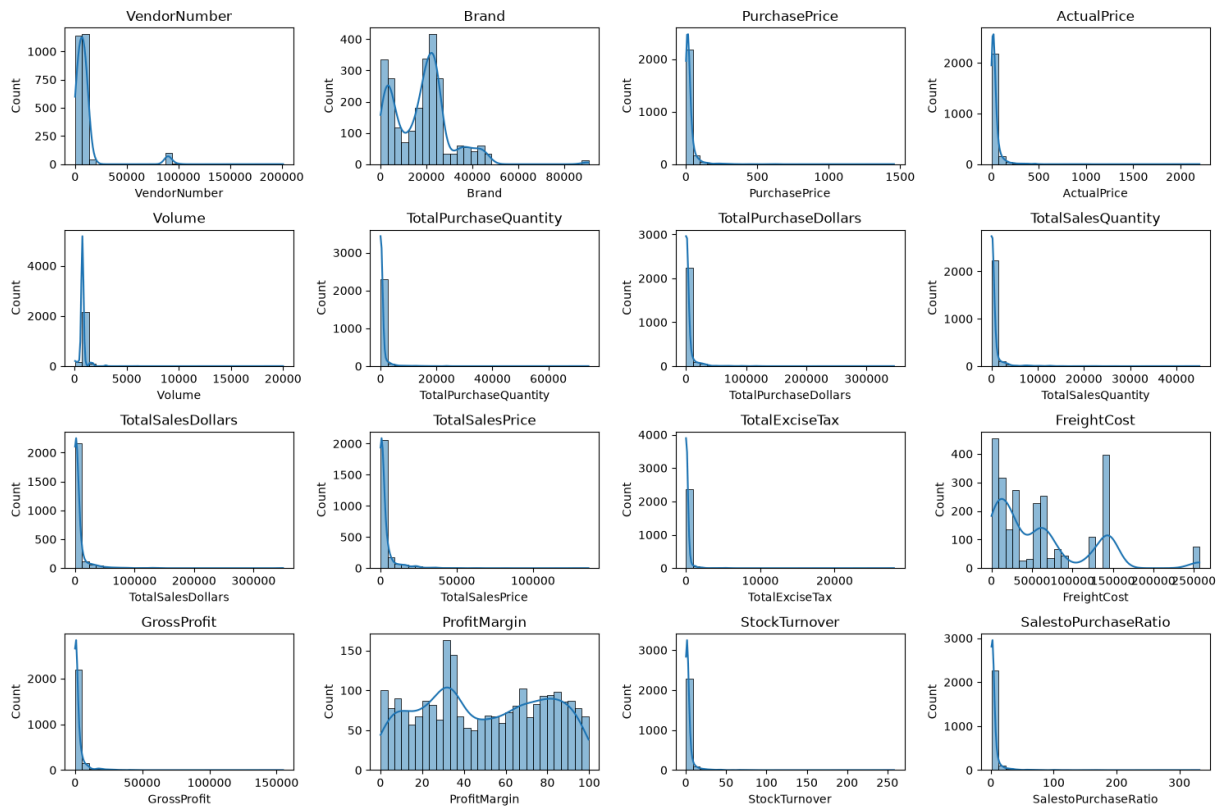
	VendorNumber	VendorName	Brand	Description	PurchasePrice	ActualPrice
0	4425	MARTIGNETTI COMPANIES	23143	Bogle Chard	4.65	6.79
1	1128	BROWN-FORMAN CORP	5299	Southern Comfort	15.26	21.99
2	12546	JIM BEAM BRANDS COMPANY	1476	Jim Beam Traveler	16.05	21.99
3	2000	SOUTHERN WINE & SPIRITS NE	11970	Zaccagnini Montepulciano Rsv	7.71	18.99
4	6785	PALM BAY INTERNATIONAL INC	8746	Cavit Pinot Grigio	4.02	6.99
...	...	...	...	...	...	...
2437	3960	DIAGEO NORTH AMERICA INC	2626	Crown Royal Apple	1.42	1.99
2438	9815	WINE GROUP INC	8527	Concannon Glen Ellen Wh Zin	1.32	4.99
2439	8004	SAZERAC CO INC	5683	Dr McGillicuddy's Apple Pie	0.39	0.49
2440	3960	DIAGEO NORTH AMERICA INC	6127	The Club Strawbry Margarita	1.47	1.99
2441	7245	PROXIMO SPIRITS INC.	3065	Three Olives Grape Vodka	0.71	0.99

2442 rows × 18 columns

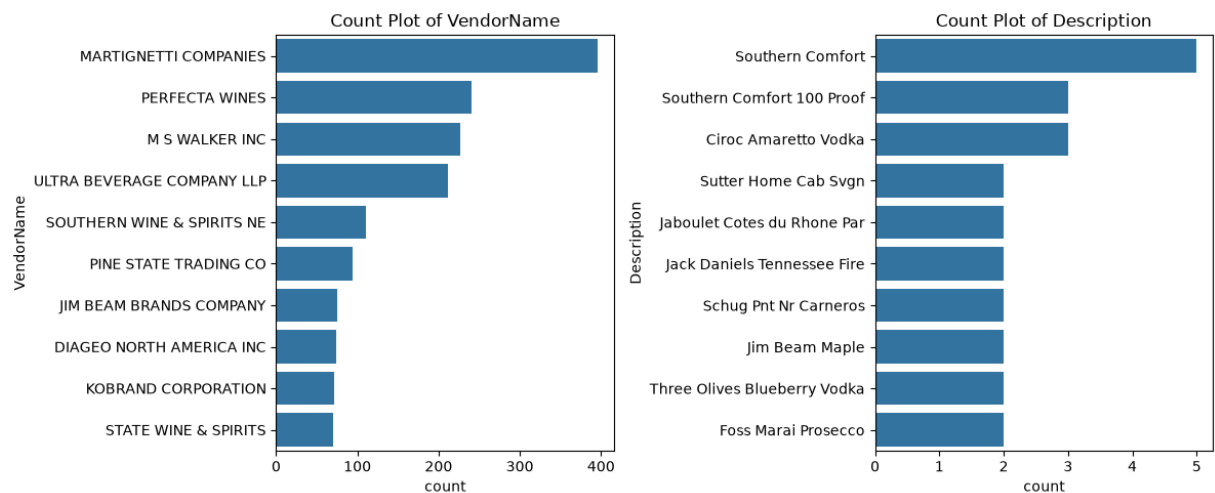


In [12]:

```
# Distribution Plots for Numerical Columns
numerical_cols = df.select_dtypes(include=np.number).columns
plt.figure(figsize=(15, 10))
for i, col in enumerate(numerical_cols):
    plt.subplot(4, 4, i + 1)      # Creates a 4 x 4 grid of subplots
    sns.histplot(df[col], kde=True, bins=30)
    plt.title(col)
plt.tight_layout()
plt.show()
```

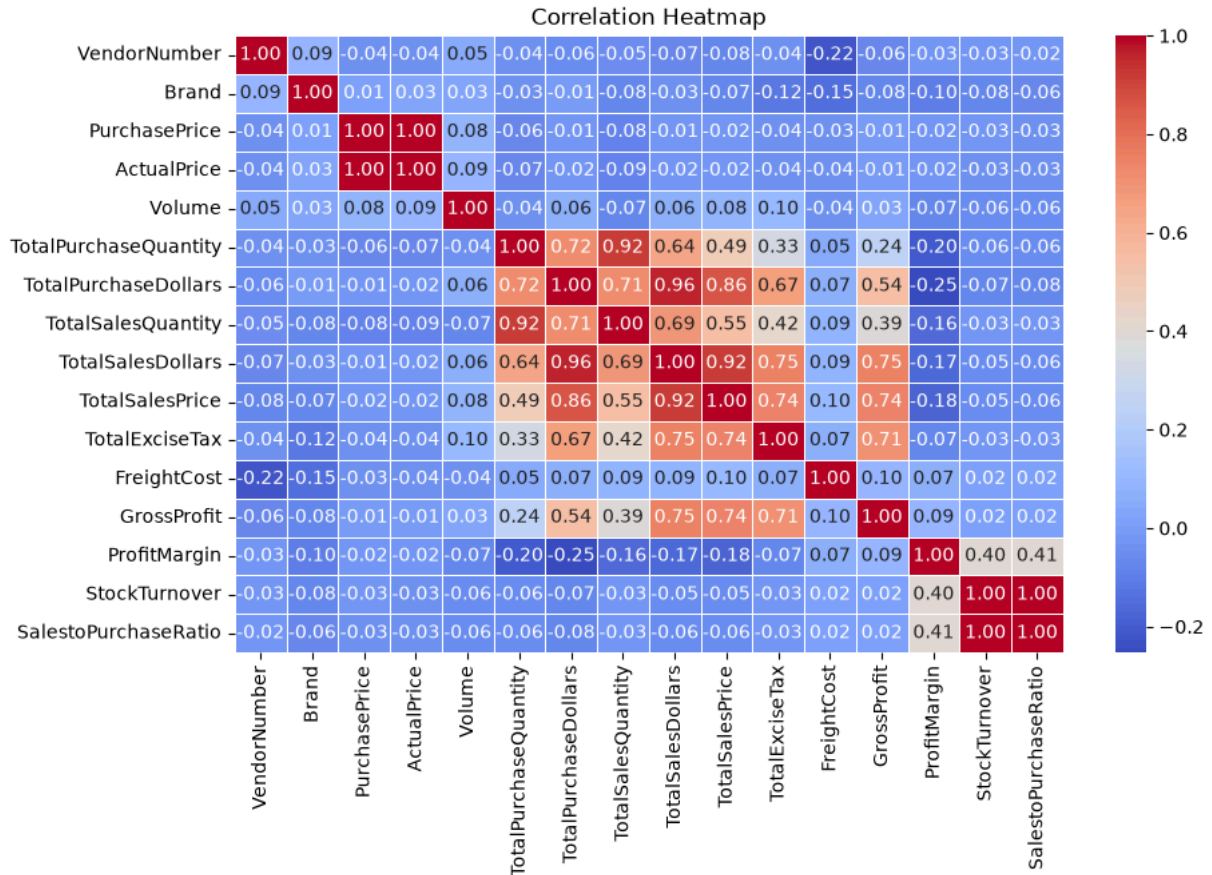


```
In [13]: # Count Plots for Categorical Columns
categorical_cols = ["VendorName", "Description"]
plt.figure(figsize=(12, 5))
for i, col in enumerate(categorical_cols):
    plt.subplot(1, 2, i + 1)
    sns.countplot(y=df[col], order=df[col].value_counts().index[:10]) # Top 10 cat
    plt.title(f"Count Plot of {col}")
plt.tight_layout()
plt.show()
```



```
In [21]: # Correlation Heatmap
plt.figure(figsize=(10, 6))
correlation_matrix = df[numerical_cols].corr()
sns.heatmap(correlation_matrix, annot=True, fmt=".2f", cmap="coolwarm", linewidths=
```

```
plt.title("Correlation Heatmap")
plt.show()
```



CORRELATION INSIGHTS 1.PurchasePrice and ActualPrice have a perfect positive correlation (1.00), indicating that products with higher purchase prices also have proportionally higher actual selling prices. 2.TotalPurchaseDollars and TotalSalesDollars show a very strong positive correlation (0.96), suggesting that higher purchase investments generally result in higher sales revenue. 3.TotalPurchaseQuantity and TotalSalesQuantity have a strong positive correlation (0.92), indicating that vendors purchasing larger quantities tend to sell more products. 4.GrossProfit has a strong positive correlation with TotalSalesDollars (0.75) and TotalSalesPrice (0.74), showing that increased sales revenue is associated with higher gross profit.

```
In [28]: #. To identify such brands, you can filter the data based on:Low TotalSalesDollars
brand_performance=df.groupby('Description').agg({
    'TotalSalesDollars': 'sum',
    'ProfitMargin': 'mean'
}).reset_index()
```

```
In [27]: low_sales_threshold = brand_performance['TotalSalesDollars'].quantile(0.15)
high_margin_threshold = brand_performance['ProfitMargin'].quantile(0.85)
```

```
In [29]: low_sales_threshold
```

```
Out[29]: np.float64(255.70200000000003)
```

```
In [30]: high_margin_threshold
```

```
Out[30]: np.float64(84.44091506054933)
```

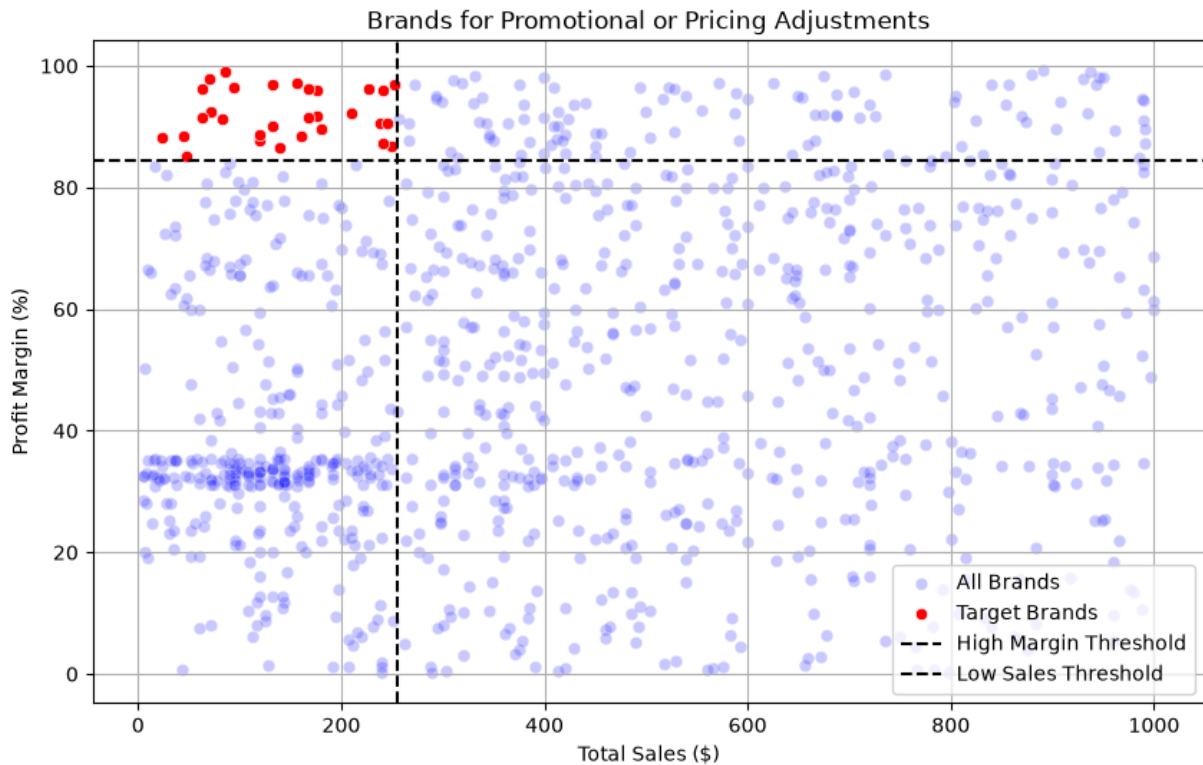
```
In [31]: # Filter brands with low sales but high profit margins
target_brands = brand_performance[
    (brand_performance['TotalSalesDollars'] <= low_sales_threshold) &
    (brand_performance['ProfitMargin'] >= high_margin_threshold)]
print("Brands with Low Sales but High Profit Margins:")
display(target_brands.sort_values('TotalSalesDollars'))
```

Brands with Low Sales but High Profit Margins:

	Description	TotalSalesDollars	ProfitMargin
776	Crown Royal Apple	23.88	88.107203
2232	Tracia Syrah	44.94	88.495772
189	Basilica Amaretto	47.45	85.079031
879	Dr McGillicuddy's Apple Pie	62.72	96.269133
1923	Sauza Sparkling Margarita	62.91	91.448100
2167	The Club Strawbry Margarita	69.65	97.889447
1784	Promesses de France CdRhône	71.91	92.337644
2166	The Club Mudslide	83.58	91.325676
2193	Three Olives Grape Vodka	85.14	99.166079
105	Aresti Pnt Nr Curico Vly	94.81	96.540449
27	Absolut Orient Apple Vodka	119.94	87.652159
2309	Vigna Magana Dignus	119.94	88.736035
1847	Riunite Sweet White	131.78	96.888754
1607	Nyers Vyds Sage Canyon Red	132.93	90.145189
518	Ch La Gordonne Chapelle Rose	139.95	86.488031
1717	Piehole Apple Pie	156.42	97.238205
1349	La Petite Maison Ventoux Bl	160.77	88.555079
1452	Mad Dogs & Englishmen Jumil	167.88	96.110317
1679	Pazo de Galegos Albarino	167.92	91.609100
801	DeKuyper Buttershots Trav	175.84	95.939490
318	C Y Toro Carmenere Terrunyo	175.92	91.774670
1096	Hacienda Haro Lopez Cri Rsva	179.88	89.626418
1036	Gifft Chardonnay	209.85	92.327853
2279	Vampire Transylvania Merlot	227.15	96.240370
671	Chevalier L'Aujardiere Chard	237.86	90.599512
2388	Zhenka Vodka 80 Proof	240.39	87.303964
1546	Mojoshot Blue Lagoon RTD	241.56	96.071369
1225	Jadot Marsannay Clos Roy 12	244.93	90.601396
750	Copain Monument Tree Pnt Nr	249.95	86.841368
802	DeKuyper Mixed Berry Medley	252.77	96.850892

```
In [39]: brand_performance = brand_performance[brand_performance['TotalSalesDollars'] < 1000]
```

```
In [40]: plt.figure(figsize=(10, 6))
sns.scatterplot(
    data=brand_performance,
    x='TotalSalesDollars',
    y='ProfitMargin',
    color='blue',
    label='All Brands',
    alpha=0.2
)
sns.scatterplot(
    data=target_brands,
    x='TotalSalesDollars',
    y='ProfitMargin',
    color='red',
    label='Target Brands'
)
plt.axhline(
    high_margin_threshold,
    linestyle='--',
    color='black',
    label='High Margin Threshold'
)
plt.axvline(
    low_sales_threshold,
    linestyle='--',
    color='black',
    label='Low Sales Threshold'
)
plt.xlabel("Total Sales ($)")
plt.ylabel("Profit Margin (%)")
plt.title("Brands for Promotional or Pricing Adjustments")
plt.legend()
plt.grid(True)
plt.show()
```



```
In [44]: def format_dollars(value):
         if value >= 1_000_000:
             return f"{value / 1_000_000:.2f}M"
         elif value >= 1_000:
             return f"{value / 1_000:.2f}K"
         else:
             return str(value)
```

```
In [41]: # Top Vendors & Brands by Sales Performance
top_vendors = df.groupby("VendorName")["TotalSalesDollars"].sum().nlargest(10)
top_brands = df.groupby("Description")["TotalSalesDollars"].sum().nlargest(10)
top_vendors
```

```
Out[41]: VendorName
MARTIGNETTI COMPANIES      3208664.60
ULTRA BEVERAGE COMPANY LLP 1674543.31
BROWN-FORMAN CORP         1069929.43
PERFECTA WINES             997745.88
JIM BEAM BRANDS COMPANY    797197.58
M S WALKER INC             706501.83
DIAGEO NORTH AMERICA INC   684771.24
SOUTHERN WINE & SPIRITS NE  634401.82
BACARDI USA INC            571420.71
CONSTELLATION BRANDS INC   446948.46
Name: TotalSalesDollars, dtype: float64
```

```
In [42]: top_brands
```

```
Out[42]: Description
Southern Comfort          600063.27
Bogle Chard               349552.23
Jim Beam Traveler         340530.32
Zaccagnini Montepulciano Rsv 143439.38
Bacardi 151 Proof         137571.18
Cavit Pinot Grigio        132730.13
Capt Morgan White        131454.42
Southern Comfort 100 Proof 130774.88
Bacardi Gold Rum Traveler 127642.70
Ca'Momi Merlot Napa Valley 126733.26
Name: TotalSalesDollars, dtype: float64
```

```
In [45]: top_brands.apply(lambda x :format_dollars(x))
```

```
Out[45]: Description
Southern Comfort          600.06K
Bogle Chard               349.55K
Jim Beam Traveler         340.53K
Zaccagnini Montepulciano Rsv 143.44K
Bacardi 151 Proof         137.57K
Cavit Pinot Grigio        132.73K
Capt Morgan White        131.45K
Southern Comfort 100 Proof 130.77K
Bacardi Gold Rum Traveler 127.64K
Ca'Momi Merlot Napa Valley 126.73K
Name: TotalSalesDollars, dtype: str
```

```
plt.figure(figsize=(15, 5))
```

Plot for Top Vendors

```
plt.subplot(1, 2, 1) ax1 = sns.barplot(y=top_vendors.index, x=top_vendors.values,
palette="Blues_r" ) plt.title("Top 10 Vendors by Sales") for bar in ax1.patches: ax1.text(
bar.get_width() + (bar.get_width() * 0.02), bar.get_y() + bar.get_height() / 2,
format_dollars(bar.get_width()), ha='left', va='center', fontsize=10, color='black' )
```

Plot for Top Brands

```
plt.subplot(1, 2, 2) ax2 = sns.barplot(y=top_brands.index.astype(str), x=top_brands.values,
palette="Reds_r" ) plt.title("Top 10 Brands by Sales") for bar in ax2.patches: ax2.text(
bar.get_width() + (bar.get_width() * 0.02), bar.get_y() + bar.get_height() / 2,
format_dollars(bar.get_width()), ha='left', va='center', fontsize=10, color='black' )
plt.tight_layout() plt.show()
```

```
In [ ]: Which vendors contribute the most to total purchase dollars?
```

```
In [52]: vendor_performance=df.groupby('VendorName').agg({
    'TotalPurchaseDollars': 'sum',
    'GrossProfit': 'sum',
    'TotalSalesDollars': 'sum'
})
```

```
In [58]: vendor_performance['PurchaseContribution%'] = (vendor_performance['TotalPurchaseDollars']/
    round(vendor_performance.sort_values('PurchaseContribution%',ascending=False),2))
```

Out[58]:

	TotalPurchaseDollars	GrossProfit	TotalSalesDollars	PurchaseContribution%
VendorName				
MARTIGNETTI COMPANIES	2099482.56	1109182.04	3208664.60	21.0
ULTRA BEVERAGE COMPANY LLP	1014476.38	660066.93	1674543.31	10.1
BROWN-FORMAN CORP	725586.31	344343.12	1069929.43	7.2
PERFECTA WINES	699743.06	298002.82	997745.88	7.0
SOUTHERN WINE & SPIRITS NE	472113.01	162288.81	634401.82	4.7
...	...	...	...	
MARSALLE COMPANY	78.78	2421.83	2500.61	0.0
SILVER MOUNTAIN CIDERS	77.18	251.35	328.53	0.0
CAPSTONE INTERNATIONAL	54.64	192.23	246.87	0.0
DUGGANS DISTILLED PRODUCTS	34.77	179.10	213.87	0.0
FLAVOR ESSENCE INC	17.00	7.99	24.99	0.0

91 rows × 4 columns

```
In [61]: top_vendors.head()
```

```
Out[61]: VendorName
MARTIGNETTI COMPANIES      3208664.60
ULTRA BEVERAGE COMPANY LLP  1674543.31
BROWN-FORMAN CORP          1069929.43
PERFECTA WINES              997745.88
JIM BEAM BRANDS COMPANY     797197.58
Name: TotalSalesDollars, dtype: float64
```

```
In [65]: top_vendors = top_vendors.reset_index()
print(top_vendors.columns)
```

```
Index(['level_0', 'index', 'VendorName', 'TotalSalesDollars'], dtype='str')
```

```
In [67]: vendor_performance = df.groupby('VendorName').agg({
        'TotalPurchaseDollars': 'sum'
    }).reset_index()
vendor_performance
```

```
Out[67]:
```

	VendorName	TotalPurchaseDollars
0	ALISA CARR BEVERAGES	21370.50
1	ATLANTIC IMPORTING COMPANY	3230.68
2	BACARDI USA INC	324928.95
3	BANFI PRODUCTS CORP	176837.23
4	BLACK PRINCE DISTILLERY INC	4448.78
...	...	...
86	VRANKEN AMERICA	25059.67
87	WEIN BAUER INC	95.04
88	WESTERN SPIRITS BEVERAGE CO	18470.74
89	WILLIAM GRANT & SONS INC	24397.14
90	WINE GROUP INC	51579.09

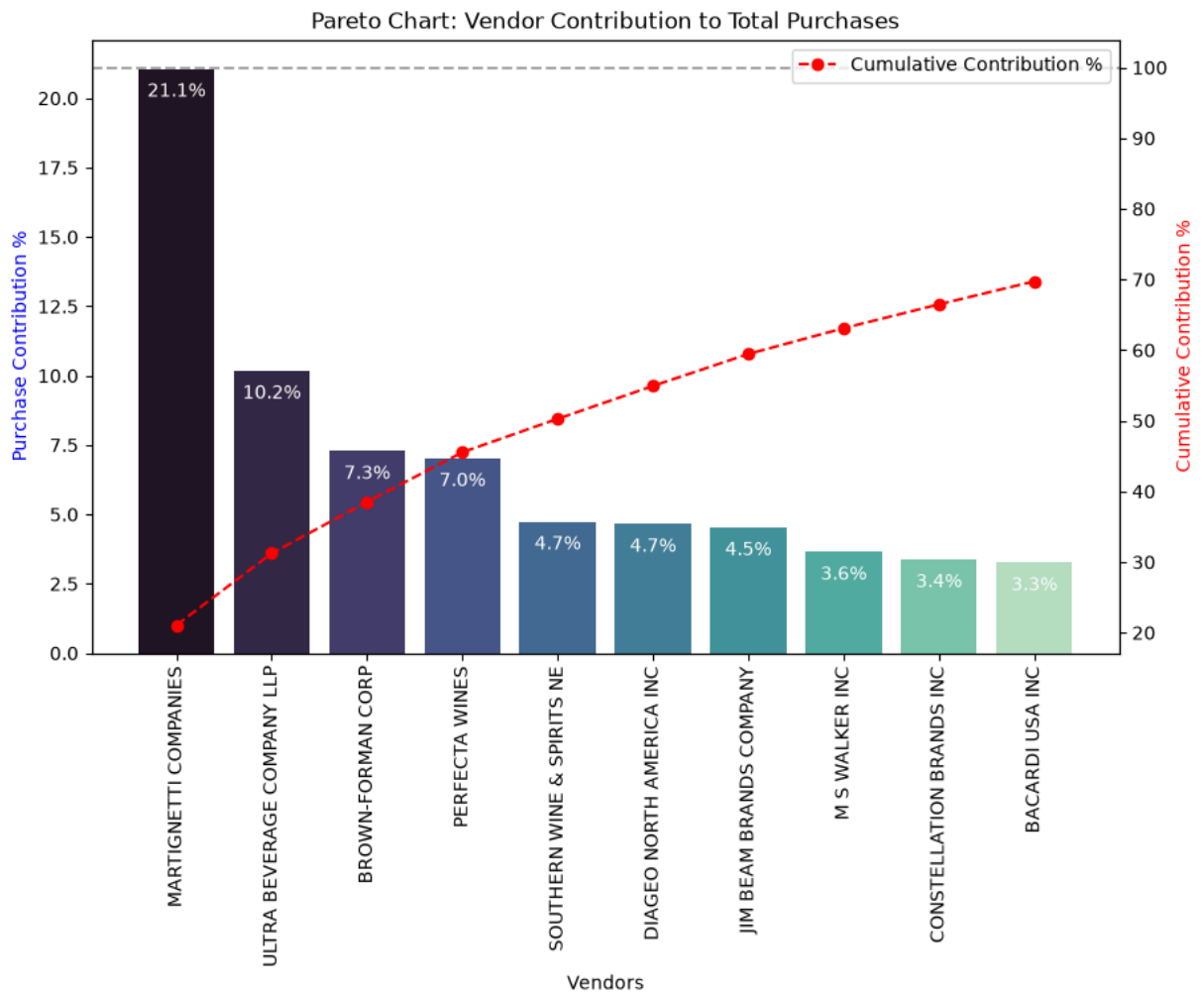
91 rows × 2 columns

```
In [71]: vendor_performance = df.groupby('VendorName').agg({
        'TotalPurchaseDollars': 'sum'
    }).reset_index()
vendor_performance['Purchase_Contribution%'] = (
    vendor_performance['TotalPurchaseDollars'] /
    vendor_performance['TotalPurchaseDollars'].sum()
) * 100
vendor_performance = vendor_performance.sort_values(
    'Purchase_Contribution%',
    ascending=False
)
vendor_performance['Cumulative_Contribution%'] = (
    vendor_performance['Purchase_Contribution%'].cumsum()
```

```
)  
top_vendors = vendor_performance.head(10)  
top_vendors.columns
```

```
Out[71]: Index(['VendorName', 'TotalPurchaseDollars', 'Purchase_Contribution%',  
              'Cumulative_Contribution%'],  
              dtype='str')
```

```
In [72]: fig, ax1 = plt.subplots(figsize=(10, 6))  
# Bar Plot for Purchase Contribution %  
sns.barplot(  
    x=top_vendors['VendorName'],  
    y=top_vendors['Purchase_Contribution%'],  
    palette="mako",  
    ax=ax1  
)  
for i, value in enumerate(top_vendors['Purchase_Contribution%']):  
    ax1.text(  
        i,  
        value - 1,  
        str(round(value, 1)) + "%",  
        ha='center',  
        fontsize=10,  
        color='white'  
    )  
# Line Plot for Cumulative Contribution %  
ax2 = ax1.twinx()  
ax2.plot(  
    top_vendors['VendorName'],  
    top_vendors['Cumulative_Contribution%'],  
    color='red',  
    marker='o',  
    linestyle='dashed',  
    label='Cumulative Contribution %'  
)  
ax1.set_xticklabels(top_vendors['VendorName'], rotation=90)  
ax1.set_ylabel('Purchase Contribution %', color='blue')  
ax2.set_ylabel('Cumulative Contribution %', color='red')  
ax1.set_xlabel('Vendors')  
ax1.set_title('Pareto Chart: Vendor Contribution to Total Purchases')  
ax2.axhline(y=100, color='gray', linestyle='dashed', alpha=0.7)  
ax2.legend(loc='upper right')  
plt.show()
```



```
In [74]: low_inventory_vendors = df[df['StockTurnover'] < 1].groupby('VendorName')[['StockTurnover', 'InventoryValue']]
low_inventory_vendors
```

Out[74]:

StockTurnover

VendorName	
WINE GROUP INC	0.535899
THE IMPORTED GRAPE LLC	0.544246
SURVILLE ENTERPRISES CORP	0.640918
ALISA CARR BEVERAGES	0.666667
TY KU LLC	0.666667
...	...
SIDNEY FRANK IMPORTING CO	0.947293
LUXCO INC	0.968750
FABRIZIA SPIRITS LLC	0.976675
REMY COINTREAU USA INC	0.976744
MCCORMICK DISTILLING CO	0.994819

67 rows × 1 columns

```
In [77]: print("Total Purchase Quantity:", df["TotalPurchaseQuantity"].sum())
print("Total Sales Quantity:", df["TotalSalesQuantity"].sum())
```

Total Purchase Quantity: 1475197.0

Total Sales Quantity: 1397551.0

```
In [79]: df["UnsoldInventoryValue"] = (
    (df["TotalPurchaseQuantity"] - df["TotalSalesQuantity"])
    .clip(lower=0)
    * df["PurchasePrice"]
)
df['UnsoldInventoryValue']
```

```
Out[79]: 0      136314.75
1         0.00
2         0.00
3      62466.42
4      45385.80
...
2437        0.00
2438        0.00
2439        0.00
2440        0.00
2441        0.00
```

Name: UnsoldInventoryValue, Length: 2442, dtype: float64

```
In [80]: # Aggregate Capital Locked per Vendor
inventory_value_per_vendor = (
    df.groupby("VendorName")["UnsoldInventoryValue"]
    .sum()
```

```

        .reset_index()
    )
    # Sort Vendors with the Highest Locked Capital
    inventory_value_per_vendor = inventory_value_per_vendor.sort_values(
        by="UnsoldInventoryValue",
        ascending=False
    )
    inventory_value_per_vendor["UnsoldInventoryValue"] = (
        inventory_value_per_vendor["UnsoldInventoryValue"]
        .apply(format_dollars)
    )
    inventory_value_per_vendor.head(10)

```

Out[80]:

	VendorName	UnsoldInventoryValue
41	MARTIGNETTI COMPANIES	405.39K
81	ULTRA BEVERAGE COMPANY LLP	158.80K
51	PERFECTA WINES	147.20K
68	SOUTHERN WINE & SPIRITS NE	145.10K
11	CONSTELLATION BRANDS INC	99.30K
79	TRINCHERO FAMILY ESTATES	85.26K
54	PINE STATE TRADING CO	68.17K
20	E & J GALLO WINERY	66.26K
78	TREASURY WINE ESTATES	62.76K
38	M S WALKER INC	59.03K

In []: